BLA-05 · ANALYTICS

Making Sense of Data

Data at rest vs data in motion · Amazon Athena and EMR · Amazon QuickSight · Amazon Kinesis and MSK · How they fit around an AI app

PART 01

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Data at rest vs data in motion

	Data at rest	Data in motion
What it is	Stored files and tables	A constant flow of events
Example	Last month's chatbot logs in S3	Clicks and messages as they happen
Question	"What happened?"	"What is happening right now?"
Services	Athena · EMR · QuickSight	Kinesis · MSK

💡 **Sort the services by one question first:** is the data sitting still, or moving?

Amazon Athena

Ask questions of files in S3 using SQL — no database to set up.

Tip	Why it saves money
Store data as Parquet	Athena reads only the columns you ask for
Partition by date	Athena skips the days you don't need

💰 You pay for how much data each query scans.

> 🏢 **"Which questions did our chatbot fail to answer most often last week?"**
> One SQL query over the logs in S3 — and it tells you what's missing from the knowledge base.

Amazon EMR

Big-data processing with Apache Spark and Hadoop — on a cluster AWS manages.

Node	Job
Primary	Coordinates the cluster
Core	Runs tasks and stores data
Task	Extra processing only — a good place for cheaper Spot instances

🏢 **Cleaning and de-duplicating 5 TB of documents** before they are embedded for a knowledge base. EMR Serverless removes the cluster management too.

Amazon QuickSight

Dashboards and charts for people who don't write SQL.

- 📊 Interactive dashboards, shared across the business
- ⚡ SPICE — an in-memory engine that keeps dashboards fast
- 💬 Ask questions in plain English with Amazon Q in QuickSight

> 🏢 **A dashboard of AI token spend per team, per week —** so managers can see AI costs without asking an engineer.

Amazon Kinesis Data Streams

A real-time stream of events that several apps can read at the same time.

Idea	What it means
Shard	A unit of capacity — more shards, more throughput
Partition key	Decides which shard each record goes to
Retention	Records stay for 24 hours by default — up to 365 days

⚠️ **Same partition key on every record** → one shard does all the work (a "hot shard"). Use a key with many values, like a user or session ID.

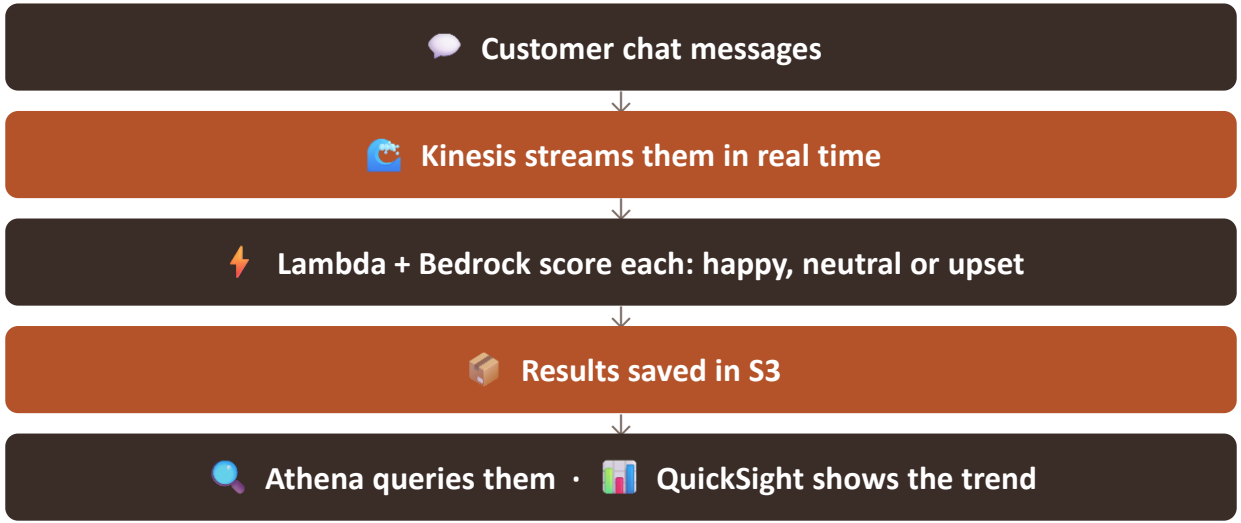
Kinesis or MSK?

	Amazon Kinesis Data Streams	Amazon MSK
What it is	AWS's own streaming service	Managed Apache Kafka
Best when	You're starting fresh on AWS	Your team already uses Kafka
Effort	Very little to run	More settings to manage

💡 **They solve the same problem.** The deciding question is usually: "Do we already use Kafka?"

Putting it around an AI app

Real-time customer mood, from chat message to dashboard.



💡 **The AI is only one step** — streaming, storage and dashboards turn its answer into action.

Key takeaways

1. First ask: is the data at rest or in motion?
2. Athena — SQL on S3, pay per data scanned
3. EMR — big transformations with Spark
4. QuickSight — dashboards for non-engineers
5. Kinesis or MSK — real-time streams; watch your partition key

Next · Part 2: where your app runs, and where it remembers